PROJECT REPORT

AI Voice Assistant (Jarvis)

Project Documentation

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1. Problem Statement

In daily life, users often need to perform repetitive tasks such as searching for information, opening websites, checking time, and sending emails. These tasks require manual effort and time. There is a need for a system that can understand voice commands and perform actions automatically.

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2. Why This Problem Matters

- Saves time and effort
- Improves productivity
- Helpful for people with physical disabilities
- Makes human-computer interaction more natural

Voice assistants are widely used today, such as Google Assistant and Amazon Alexa, which shows the importance of such systems.

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3. Proposed Solution

We developed an AI Voice Assistant (Jarvis) that can:

- Listen to user voice commands
- Convert speech into text
- Perform tasks based on commands
- Respond using voice

The system acts as a virtual assistant to help users perform daily activities efficiently.

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4. Approach / Methodology

Working Process:

1. Voice Input
   The user speaks through a microphone
2. Speech Recognition
   The system converts voice into text using a speech recognition model
3. Command Processing
   The program checks the input for specific keywords

4. Task Execution
   The corresponding function is executed
5. Voice Output
   The assistant responds using text-to-speech

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5. Technologies Used

- Python (Programming Language)
- pyttsx3 (Text-to-Speech)
- speechRecognition (Speech-to-Text)
- Wikipedia API (Information retrieval)
- Webbrowser module (Open websites)
- SMTP (Email sending)

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6. Key Features

- Voice-controlled interaction
- Wikipedia search
- Opening websites such as YouTube, Google, and others
- Playing music from the system
- Telling current time
- Sending emails

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7. Key Decisions Made

- Used pyttsx3 for offline voice output
- Used Google Speech Recognition for better accuracy
- Implemented keyword-based command system for simplicity
- Structured the program into reusable functions

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8. Challenges Faced

1. Speech Recognition Errors
   Background noise affects accuracy
2. Internet Dependency
   Speech recognition requires internet

3. Email Security
   Storing email credentials in code is unsafe
4. System Compatibility
   Voice engine works mainly on Windows

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9. Solutions to Challenges

- Adjusted pause threshold to improve recognition
- Used try-except blocks for error handling
- Recommended using secure app passwords for email
- Tested system in controlled environment

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10. Results

- The assistant successfully performs voice-based tasks
- Provides quick responses
- Makes user interaction simple and efficient

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11. What I Learned

- Basics of speech recognition systems
- Working of text-to-speech technology
- Integration of multiple Python libraries
- Handling real-time input and errors
- Designing user-friendly systems

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12. Future Improvements

- Add AI-based chatbot for smarter responses
- Integrate APIs for weather, news, and updates
- Convert into a mobile application
- Improve voice recognition accuracy
- Enhance security for email system

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13. Conclusion

The AI Voice Assistant (Jarvis) demonstrates how artificial intelligence can be used to automate daily tasks through voice interaction. It improves efficiency and provides a natural way for users to interact with technology. The project has strong real-world relevance and scope for further development